

Nitrogen-rich carbon spheres made by a continuous spraying process for high-performance supercapacitors

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ABSTRACT

Supercapacitors have high power densities, high efficiencies, and long cycling lifetimes; however, to enable their wider use, their energy densities must be significantly improved. The design and synthesis of improved carbon materials with better capacitance, rate performance, and cycling stability has emerged as the main theme of supercapacitor research. Herein, we report a facile synthetic method to prepare nitrogen-rich carbon particles based on a continuous aerosol-spraying process. The method yields particles that have high surface areas, a uniform microporous structure, and are highly N-doped, resulting in a synergism that enables the construction of supercapacitors with high energy and power density for use in both aqueous and commercial organic electrolytes. Furthermore, we have used density functional theory calculations to show that the improved performance is due to the enhanced wettability and ion adsorption interactions at the carbon/electrolyte interface that result from nitrogen doping. These findings provide new insights into the role of heteroatom doping in the capacitance enhancement of carbon materials; in addition, our method offers an efficient route for large-scale production of doped carbon.

1 Introduction

Supercapacitors have a broad range of applications due to their high power densities, high efficiencies, and long cycling lifetimes; however, the current generation of supercapacitors still have significantly lower energy densities in comparison with commonly used batteries [1–3]. Generally, the energy density (E) of an electrochemical double-layer capacitor (EDLCs)

can be described by $E = 0.5CV^2$, where V is the voltage and C is the capacitance calculated as $1/C = 1/C_{\text{Anode}} + 1/C_{\text{Cathode}}$. For symmetric or asymmetric supercapacitors, carbon electrodes are used as both or one of the electrodes, respectively, and the capacitance is generally limited by the carbon electrodes, which often have lower capacities than other electrode materials [3–5]. Developing carbon materials with significantly improved capacitance at high rate has

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been the main theme of the field.

To date, a large variety of carbon materials have been explored for supercapacitor applications, and focus has been placed on engineering the carbon frameworks (e.g., activated carbons [6], carbon nanotubes [7], carbon fibers [8], porous graphene [9], and carbide-derived carbons [10]), tuning the pore structures (e.g., microporous carbon [11], ordered mesoporous carbon [12, 13] and hierarchical porous carbon [14]), and doping carbon materials with heteroatoms (e.g., phosphorus [15], boron [16], sulfur [17], and nitrogen [5, 18–23]). Nitrogen-doping, in particular, has attracted much attention because of the electronegativity of nitrogen and its ability to form stable N-doped frameworks [24, 25]. Nitrogen doping of carbon materials affects their surface properties and electronic structure, in particular, enhancing properties that are critical for energy storage applications. In term of supercapacitors, it has been suggested that nitrogen doping can induce faradaic redox reactions and stronger adsorption interactions between the N-substituted carbon structure and electrolyte ions, increasing the capacitance [18–21]; in addition, doping improves the hydrophilicity of the carbon materials, effectively increasing the ion-accessible surface, facilitating ion diffusion, and reducing the charge transfer resistance of the carbon electrode [5, 18, 26, 27]. Due to the advantages of N-doping in enhancing the electrochemical properties, many N-doped carbons have

been prepared either by post-treatment of carbon materials with N-containing substances or by carbonization of N-containing carbon precursors, and these materials show significant improvement in capacitance over non-modified materials [19, 22, 23, 25–28].

By allowing simultaneous control over the morphology, pore structure, and degree of N doping, we hypothesized that high-performance N-doped carbons could be prepared by a continuous aerosol-spraying process. As illustrated in Fig. 1, we started from aqueous solutions containing the carbon precursor (a copolymer of melamine, phenol, and formaldehyde, denoted as MPF) and a pore template (silicate clusters made by hydrolysis and condensation of tetraethoxy-silane). An atomization process using nitrogen carrier gas was used to continuously generated aerosol droplets. The droplets passed through a high-temperature zone, during which evaporation of the solvent promoted the co-assembly of the MPF precursors with the silicates, leading to the formation of polymer-silicate composite particles. Subsequent carbonization and removal of the silicate templates converted the particles into nitrogen-rich carbon (NRC) particles.

In our design, melamine, which has a high nitrogen content (~66.7 wt.%), was used as the co-monomers of MPF, enabling the formation of NRC particles with a high degree of N-doping. The silicate clusters can form hydrogen bonds with the MPF precursor via silanol

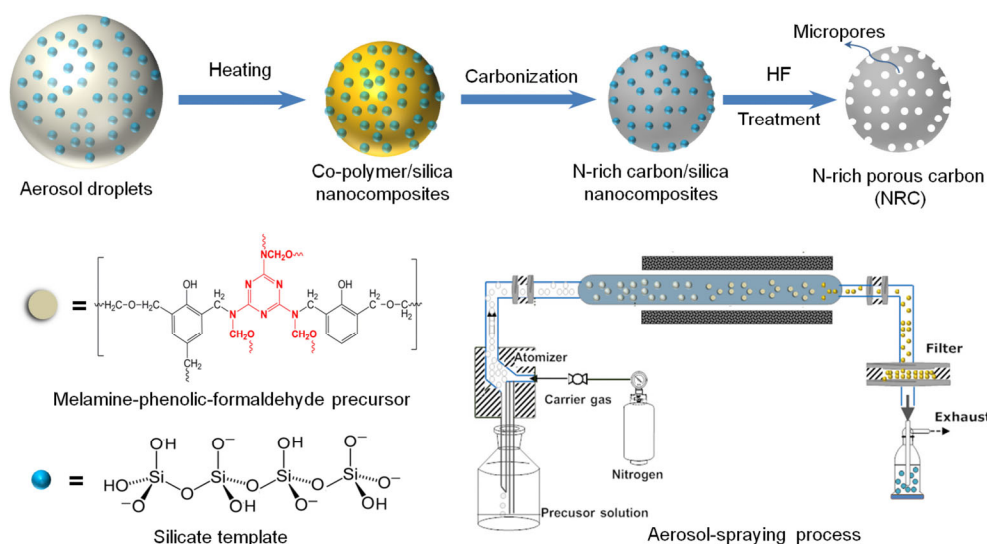


Figure 1 Schematic for the formation of aerosol-assisted NRC.

groups in the silicates and the phenol groups in MPF, enabling their co-assembly into composite particles with uniform cluster size distribution. Subsequent removal of the templates led to the formation of NRC particles with uniform pores and high surface area. The yield of NRC was calculated by correlating the total mass of feed carbon and nitrogen source with the mass of final NRC sample is approximately 11%. In addition, the as-formed NRC particles are polydisperse in size (<1 micrometer in diameter), enabling dense packing in the electrodes where packing voids allow effective electrolyte transport. Furthermore, the spraying process, exemplified herein by the aerosol process, has been widely used by industry for large-scale powder fabrication, such as the large-scale production of electrode materials for lithium-ion batteries [14, 29–31]. Thus, this synthetic strategy offers an efficient production route for NRC particles, facilitating low-cost production for commercial applications.

2 Materials and methods

2.1 Preparation of NRC, bulk N-rich carbon (bulk-NRC), and nitrogen-free carbon (NFC) samples

For the synthesis of the NRC particles, melamine (1.2 g) and phenol (0.6 g) were added to a round-bottomed flask and melted at 40 °C. Then, deionized water (10 mL) and 1 M NaOH (0.4 mL) solution were sequentially added into the round-bottomed flask to adjust the pH of the solution to 9. Subsequently, formalin (10 mL, 37 wt.%) was added dropwise to the mixture, and the reaction mixture was stirred and refluxed at 80 °C for 1 h. Then, a solution containing tetraethoxysilane (TEOS, 2.1 g), ethanol (10 mL), 0.1 M aqueous HCl (1 mL), and deionized water (10 mL) was added to the reaction system, and it was rapidly cooled to room temperature. After stirring for a further 30 min, a homogeneous precursor solution was obtained. The precursor solution was atomized using nitrogen as the carrier gas, and the heating zone temperature was maintained at 450 °C. The as-collected particles were carbonized at 900 °C under an N₂ atmosphere for 180 min at a heating rate of 8 °C·min⁻¹. The carbonized particles were washed using 10 wt.% HF and deionized water

to yield the NRC samples. For the synthesis of the bulk NRC sample, the precursor solution was directly dried at 100 °C in an oven to obtain the bulk powder, which was then subjected to the same carbonization and washing process. For the preparation of the NFC samples, melamine-free precursor solution was prepared using a similar procedure, and the particles were synthesized using the similar process.

2.2 Material characterization

Scanning electron microscopy (SEM, JSM-7401F) and transmission electron microscopy (TEM, JEOL-2010; Tecnai G2F20S-Twin) were used to visualize the microstructure of prepared samples. The pore structure was determined by N₂ sorption at -196 °C using an ASAP 2020 volumetric sorption analyzer. Brunauer–Emmett–Teller (BET) surface areas were calculated from the isotherm using the BET equation. The pore size distribution was calculated using nonlocal density functional theory (NLDFT) method on the basis of the adsorption branch. X-ray diffraction (XRD) measurements were made using a Rigaku D/Max 2400 diffractometer and CuK α radiation (40 kV, 100 mA, λ = 1.5406 Å). Raman spectroscopy measurements were made with a Renishaw inVia Micro-Raman spectrometer at 532 nm. X-ray photoelectron spectroscopy (XPS) was performed on a PHI 5700 ESCA System using AlK α X-ray at 14 kV and 6 mA. Binding energies (BE) were calibrated with respect to C1s peak at 284.6 eV. Thermogravimetry analysis (TGA Q50 instruments) was performed at a heating rate of 10 °C·min⁻¹ under an Ar atmosphere. Conductivity measurements were conducted on SZT200 conductivity test device.

2.3 Fabrication of electrodes and supercapacitor

The active material (80 wt.%), carbon black (10 wt.%), and polytetrafluoroethylene (PTFE) (10 wt.% of a 60 wt.% dispersion in water) were homogeneously mixed and rolled into a uniform thin film. The above thin film was dried at 70 °C for 8 h and then tailored into ~1 cm² electrodes for further treatment. The loading of the active material on each electrode was ~1.0 mg. In the three-electrode test system, the thin film electrode was pressed onto a nickel foam current collector for use as the working electrode. Pt foil and a saturated

calomel electrode (SCE) were used as the counter electrode and reference electrodes, respectively. The electrochemical performance was tested in 6 M KOH aqueous electrolyte at room temperature. The symmetric supercapacitors were constructed by using aqueous (6 M KOH) or organic (1 M tetraethylammonium tetrafluoroborate in propylene carbonate (TEABF₄-PC)) electrolytes. For the organic supercapacitors, the electrode was pressed onto conductive carbon coated aluminum foils (current collector) and dried at 120 °C for 12 h to remove the residual moisture and solvent. Then, two identical (by weight and size) electrodes were assembled in a CR2032 stainless steel coin cell with the glass fiber membrane (GF/D) (Whatman) as the separator. The assembly of the coin cells was performed in a glove box filled with Ar.

2.4 Electrochemical measurements

A VMP3 Electrochemical Workstation (Bio-logic) was used for electrochemical measurements. Electrochemical impedance spectroscopy was performed using a Solartron 1287/1260 electrochemical workstation at frequencies ranging from 100 kHz to 10 MHz at an amplitude of 10 mV. The specific gravimetric capacitances (C_s , F·g⁻¹) of three-electrode system were calculated based on the following formula $C_s = I\Delta t/(m\Delta V)$, where I is the constant discharge current, Δt is the time for a full discharge, m is the mass of active material in one electrode, and ΔV is voltage drop on discharge (excluding the internal resistance (IR) drop). The specific capacitances of the symmetric supercapacitors were calculated using the following equation $C_s = 2I\Delta t/(m\Delta V)$. The energy density of the cells was calculated by $E = C_s V^2/8$, where V is the cell operating voltage (obtained from the discharge curve subtracted by the IR drop). The various power densities of the device (P) were calculated from the following formula $P = E/\Delta t$, where Δt is the discharge time.

2.5 Computational method

Calculations were carried out using density functional theory (DFT) and the cluster model as implemented in Gaussian 03 [32]. We used a fused aromatic hydrocarbon structure to simulate the carbon surface. Different degrees of nitrogen doping in the carbon framework

were modeled to obtain the desired nitrogen-doping surface models, as described above. All atoms were modeled at the B3LYP/6-31G(d) level of theory. Full geometry optimization followed by frequency analysis to determine if the geometry was a stationary point on the potential energy surface were carried out for both surface and adsorption models. The adsorption energy E_{ad} is defined as $E_{ad} = E_{complex} - (E_{K^+} + E_{surface})$, where $E_{complex}$, E_{K^+} , and $E_{surface}$ represent the energies of the adsorption complex, the isolated K⁺ ion, and the carbon surface, respectively. Visual molecular dynamics (VMD) [33] was used for visualization and image preparation.

3 Results and discussion

3.1 Structural characterization

Figure 2(a) shows scanning electron microscopy images of the NRC particles, which exhibit a spherical morphology with diameters ranging from 50 to 400 nm. Figure 2(b) shows a transmission electron microscopy image of an NRC particle, confirming the spherical morphology. Figures 2(c) and 2(d) show high-resolution TEM (HRTEM) images and corresponding fast Fourier transformation (FFT) images of the NRC, confirming the amorphous microporous carbon structure with pores of ~2 nm in diameter. X-ray diffraction measurements indicate the NRC particles have an amorphous structure (see Fig. S1 in the Electronic Supplementary Material (ESM)). Figure 2(c) shows the N₂ adsorption/desorption isotherms of the NRC particles, which exhibit a typical type I adsorption isotherm, confirming the microporous structure. These micropores are uniformly distributed and the diameter distribution is centered at ~2 nm (Fig. 2(c), inset), in agreement with the TEM results. The BET surface area and pore volume of these particles are 1,519 m²·g⁻¹ and 0.81 cm³·g⁻¹, respectively. Elemental analysis of the samples indicates that the N content of these NRC particles is ~10.8 wt.%, which is much higher than that reported for N-doped microporous carbons [19]. Figure 2(d) shows the Raman spectrum of the NRC particles, and the peaks at 1,350 and 1,596 cm⁻¹ represent the D and G bands of the carbon framework, respectively [32]. The D- and the G-bands have intensities, suggesting that there



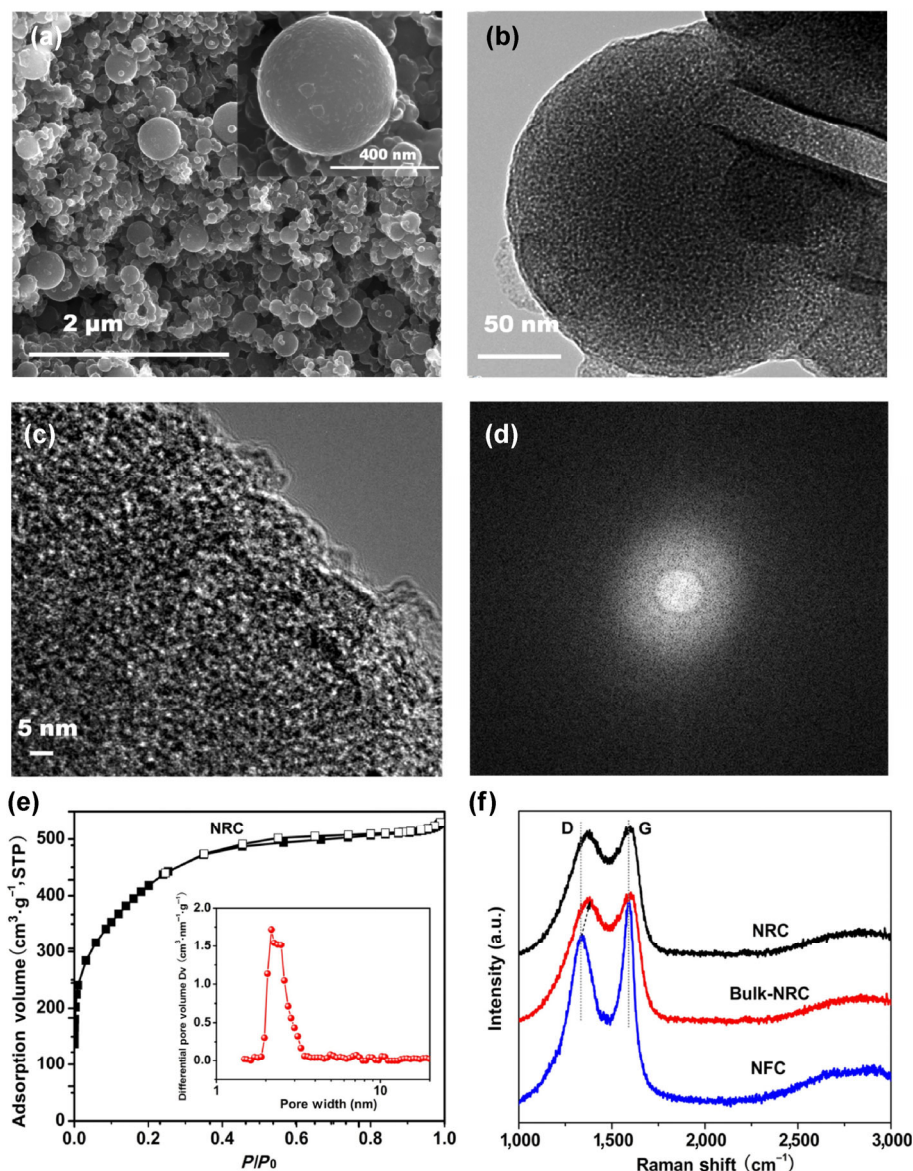


Figure 2 (a) SEM images of NRC particles. The inset shows an SEM image of an NRC particle. (b) TEM image of an NRC particle showing a microporous structure. (c) HRTEM image of NRC showing a microporous and amorphous carbon structure. (d) FFT image of NRC further revealing an amorphous structure. (e) Nitrogen sorption isotherm of the NRC particles and pore size distribution obtained from nonlocal density functional theory calculations (inset). (f) Raman spectra of NRC, NFC, and bulk-NRC.

are a significant number of structural defects in carbon framework due to N doping [34]. By changing the amount of silicate template used or the amount of melamine co-monomer used, microporous NRC particles with different surface areas, pore volumes, and N content could be readily synthesized (see Table S1 in the ESM). More specifically, as shown in Table S1 (in the ESM), maintaining the same melamine content while changing the TEOS dose of the precursor yields the resulting NRC-1, NRC-2, NRC-3, and

NRC-4 with different N contents. It is because the pre-carbonization NRC precursors originate from the co-polymerization of melamine, phenolic and MPF and the co-assembly of MPF and silicates from TEOS hydrolysis and condensation. Changing the ratio of any components in the reactions could alter the precursor structures including the number of hydrogen bonds, $-\text{CH}_2\text{O}$ micro-domains and $-\text{NCH}_2\text{O}$. Changes to these factors significantly affect the development of porosity and N doping. Therefore, there is an optimal

ratio of TEOS and melamine, and this was found to be 2.1:1.2), represented by sample NRC-3. NRC-3 has the highest BET surface area and degree of N doping.

To understand the effect of N-doping on the framework structure, we synthesized NFC from a phenolic-formaldehyde precursor (without melamine) using the same method (see the Experimental section). The as-synthesized NFC particles have a similar pore structure and surface area (see Fig. S2 in the ESM). As shown in Fig. 2(d), the Raman spectrum of the NFC particles clearly shows a higher G-to-D band ratio, indicating the NFC particles have a more ordered, crystalline structure than that of the NRC particles. Therefore, we concluded that N-doping results in a more disordered structure containing more defects. For further comparison, N-rich carbon was also synthesized by casting the MPF/silicate solution, followed by a similar carbonization process (bulk-NRC). As expected, the bulk-NRC has a similar Raman spectrum to that of the NRC particles, suggesting that the aerosol synthesis procedure has an insignificant effect on the carbon framework structure. In addition, the Raman spectra of both the NRC particles and the bulk-NRC powder contain D bands at higher frequencies than those of the NFC particles, which is consistent with the higher binding energy of the C–N bond than that of the C–C bond [34]. Overall, these results clearly indicate the successful incorporation of nitrogen atoms into the NRC particles.

To further explore the chemical environment of the

dopant, XPS was used to investigate the N species in the NRC particles (Fig. 3(a) and Fig. S4 in the ESM). The N1s spectrum can be deconvoluted into four peaks, representing the structures of pyridinic-N (N-6) at 398.5 ± 0.2 eV, pyrrolic or pyridonic-N (N-5) 399.8 ± 0.2 eV, quaternary N (N-Q) 401.5 ± 0.2 eV, and oxidized N (N-X) at 402.5 ± 0.2 eV, respectively [5, 18, 34]. Figure 3(b) illustrates the structure of these N species in a carbon plane lattice (inset), as well as their atomic percentages in the NRC particles. The N-6 and N-5 structures with unpaired electrons may induce surface polarization of the carbon frameworks, considered to be favorable to enhance the capacitance [18]. The N-Q structure serves as a component of the planar graphitic carbon, which may cause non-uniform distribution of electrons in the π -system, enhancing electron transport throughout the carbon plane. The N-X structure is a dangling oxidized structure with less contribution to the capacitance [21]. Because the former three account for the majority of the total dopant structures (>90%), it is expected that such N-doped carbons will exhibit better capacitive performance.

3.2 Electrochemical performance

The electrochemical performance of the NRC particles was examined in a three-electrode system using 6 M KOH as the electrolyte. Figure 4(a) shows the results of cyclic voltammetry (CV) studies of the NRC particles. Quasi-rectangular shape curves with minor redox

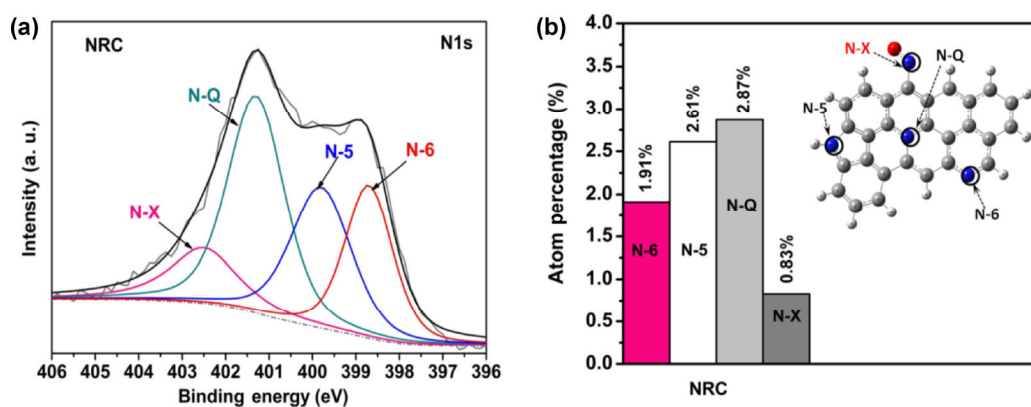


Figure 3 (a) XPS N1s spectrum of NRC deconvoluted into four sub-peaks corresponding to N-6, N-5, N-Q, and N-X structures. (b) The atomic percentages in each of the four structures. The inset shows a carbon surface model containing the four doping structures (blue, red, gray, and white balls indicate N, O, C, and H, respectively).

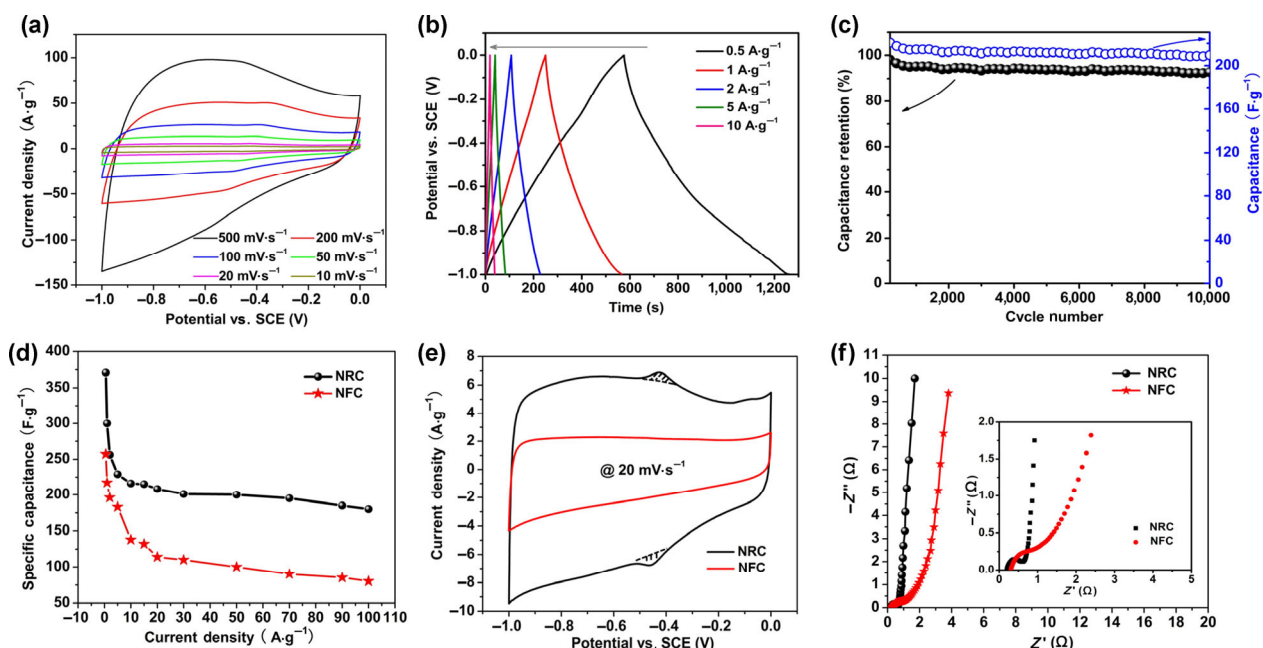


Figure 4 Electrochemical performances of the NRC and NFC electrodes in 6 M KOH electrolyte. (a) CV curves of the NRC particles at the scan rates of 10 to 500 $\text{mV}\cdot\text{s}^{-1}$. (b) Galvanostatic charge-discharge curves of the NRC electrodes at different current densities. (c) Cycling stability of the NRC electrodes at a current density of 5 $\text{A}\cdot\text{g}^{-1}$. (d) CV curves of the NRC and NFC electrodes at a scan rate 20 $\text{mV}\cdot\text{s}^{-1}$. (e) Gravimetric capacitances of NRC and NFC particles at different current densities. (f) Nyquist plots of NRC and NFC. The inset shows the expanded high-frequency region of the plots.

peaks at about -0.5 V (vs. SCE) are clearly observed at scan rates from 10 to 500 $\text{mV}\cdot\text{s}^{-1}$. Figure 4(b) and Fig. S5 (in the ESM) show the galvanostatic charge-discharge curves of the NRC particles, with distorted linear slopes and minor voltage drops. The gravimetric capacitance calculated is ~ 371 $\text{F}\cdot\text{g}^{-1}$ at 0.5 $\text{A}\cdot\text{g}^{-1}$, which is high in comparison with previously reported N-doped carbons [12, 17–20]. Even at an ultra-high current density of 100 $\text{A}\cdot\text{g}^{-1}$, the NRC particles still provide a gravimetric capacitance of 180 $\text{F}\cdot\text{g}^{-1}$. Interestingly, we found that the discharge time is longer than the corresponding charge time at low current densities of 0.5 and 1 $\text{A}\cdot\text{g}^{-1}$ (Fig. 4(b)), an observation that has also been reported previously for three-electrode systems [12, 13, 20]. This effect can be mainly ascribed to stable electrode polarization near the end point potential (i.e., around -1.0 V vs. the reference electrode), accounting for the Faradaic capacitance contribution. On increasing the current density, the polarization of the NRC electrode near the end point voltage will be diminished. In addition, the NRC particles are also very stable during cycling. The electrode has a specific capacitance of 210 $\text{F}\cdot\text{g}^{-1}$ at a current density of 5 $\text{A}\cdot\text{g}^{-1}$

and maintains more than 92% of the capacitance after 10,000 cycles (Fig. 4(c)).

Figure 4(d) shows the specific capacitance of the NRC electrode at different current densities, from which we can see that the main capacitance decay occurs at low current densities. Correlated with the longer discharge time at low rates, the relatively poor rate capability at low current densities can be attributed to the gradually decreasing contribution of faradaic capacitance due to electrode polarization near the end point potential. At high current rates (>30 $\text{A}\cdot\text{g}^{-1}$), the electrode exhibits near constant capacitance, indicating a rapid charge-discharge performance. For comparison, the non-doped NFC electrode exhibits a significantly lower capacitance, particularly at high current densities. Consistently, the NFC electrode also exhibits CV curves with less regular shapes and more IR drops than those of the NRC electrode (Figs. S6(a) and S6(b) in the ESM). These results presented clearly demonstrate that N-doping improves the capacitive performance significantly.

It has been proposed that the enhanced capacitance achieved by N-doping arises due to the extra

pseudocapacitance provided by the faradaic reactions of the N-dopants [18, 20, 21, 35]. Figure 4(e) shows the CV curves of the NRC and NFC electrodes at a scan rate $20 \text{ mV}\cdot\text{s}^{-1}$. Unlike the nitrogen-free NFC electrode, the nitrogen-doped NRC electrode has a minor redox peak at around -0.5 V , confirming that the N dopants act as redox sites. However, the contribution of pseudocapacitance (estimated from the shaded areas) to the total capacitance is less than 2%, as estimated from the entire rectangular area. Therefore, pseudocapacitance may not be the main factor contributing to the observed improved capacitance. To demonstrate that the improved capacitance is caused by N-doping alone, we further prepared two samples that have similar surface areas but that contain different amounts of doped N; we achieved this by adjusting the ratio of TEOS and melamine in the precursor. These two samples are bulk-NRC and bulk N-free carbon (bulk-NFC). Both samples have very similar surface areas, i.e., $1,093$ and $1,081 \text{ m}^2\cdot\text{g}^{-1}$, respectively (Table S2 in the ESM). By comparing the capacitive performance of bulk-NRC and bulk-NFC (Figs. S7 and S8 in the ESM), we found that the CV curve of bulk-NRC contains not only redox peaks but also a larger rectangular area than that of bulk-NFC. In addition, the increased capacitance brought by the increased rectangular area is higher than that of the redox peaks. Furthermore, the surface-area-normalized specific capacitances at $0.5 \text{ A}\cdot\text{g}^{-1}$ for bulk-NRC and bulk-NFC are ~ 25.6 and $19.9 \mu\text{F}\cdot\text{cm}^{-2}$, respectively, undoubtedly suggesting that the improved capacitance for bulk-NRC is due to N doping.

Figure 4(f) shows the Nyquist plots of the NRC and NFC electrodes in 6 M KOH . The straight line in the low-frequency region and the semicircle in the high-frequency region indicate typical capacitive behavior. The intrinsic ohmic resistance (the first intercept along the real axis) of the NRC electrode is slightly smaller than that of NFC, indicating the better electrolyte accessibility. By extrapolating the vertical portion to the real axis, the equivalent series resistance (ESR) of the NRC electrode was found to be 0.65Ω , smaller than that of NFC (1.25Ω), which confirms the lower contact resistance and better intrinsic electronic properties of NRC [5, 18]. Moreover, the straight line in the low-frequency region of NRC

electrode has an increasing slope compared to that of the NFC electrode, indicating increased ion/electron transportation. The results of electrochemical impedance spectroscopy further highlight the effect of N-doping on the ion/electron accessibility and conductivity, increasing both and improving the capacitive performance of the electrode. Although the Raman results indicate a lower ratio of graphitic structure in NRC (Fig. 2(f)), we suggest that the lack of a significantly graphitic ($C_{\text{sp}2}$) structure does not result in low conductivity because the electron transfer mechanism in planar N-doped carbon may differ from that of pure planar graphitic carbon. As a matter of fact, nitrogen doping, especially that giving rise to the graphitic N structure adds electrons and induces a non-uniform distribution of electrons in the π -system of the planar graphitic carbon, enhancing electron transport throughout the carbon plane, which has been discussed previously [18, 34]. Furthermore, we carried out conductivity tests for NRC and NFC to demonstrate the improved conductivity caused by N-doping. As expected, the electrical conductivities (σ) are $6.54 \text{ S}\cdot\text{cm}^{-1}$ for NRC and $5.95 \text{ S}\cdot\text{cm}^{-1}$ for NFC, clearly demonstrating the improved electron conductivity on N doping.

3.3 Density functional theory calculations

To further understand the effect of N doping, density functional theory calculations were performed using the cluster model for various carbon surfaces: 1) carbon surfaces containing a pyridinic, pyrrolic, or graphitic N atoms; 2) a simulated NRC surface with contents and species of N-dopant similar to those of the NRC particles; and 3) a simulated non-doped carbon (graphite) surface (Figs. S9 and S10 in the ESM). The construction of similar cluster models for simulating heteroatom-doped carbon surfaces has been used previously [36–38]. Nitrogen has a higher electronegativity than carbon [38] (their Pauling electronegativities are 3.0 and 2.5, respectively); therefore, N-doping leads to polarization of the doped carbon surface.

The DFT calculations clearly suggest that N-doping effectively induces surface polarization, as evidenced by the increased dipole moment. Dispersion forces (London dispersion force), dipole–dipole interactions

(the Keesom force), and dipole-induced-dipole interactions (the Debye force) are generally referred as the van der Waals forces, and these are the essential interactions between the carbon electrodes and electrolytes (particularly, organic electrolytes). The increased polarization results in more intense electrode/electrolyte interactions and increased wettability, which favors the transport of ions crossing the electrical double layer. In addition, the uneven electrostatic potential of the surface promotes electrostatic interactions between the dopants and the electrolyte ions (e.g., K^+ ions), leading to more accessible ion adsorption sites within the carbon surface, and consequently, enhanced capacitance.

To further explore the interactions between the electrolyte ions and the doped surface, a K^+ ion was modeled on the undoped carbon surface and the simulated NRC surface, of which the structures were optimized *in vacuo*. Figure 5(d) shows the optimized structure for the non-doped surface, where the K^+ ion is located above the centroid of the carbon π -system at a distance

of 2.768 Å from the surface; the adsorption energy for this arrangement is $-121.26 \text{ kJ}\cdot\text{mol}^{-1}$. Figures 5(e) and 5(f) illustrate the optimized structures of the simulated NRC surface, where the K^+ ion is adsorbed by either a pyridinic nitrogen (Fig. 5(e)) or a pyrrolic nitrogen (at shorter separations of 2.64 and 2.61 Å, respectively, and higher adsorption energies of -141.1 and $-180.4 \text{ kJ}\cdot\text{mol}^{-1}$, respectively. For the graphitic N sites, a shortened distance (2.678 Å) and increased adsorption energy ($-143.48 \text{ kJ}\cdot\text{mol}^{-1}$) were also found (Fig. S11(b) in the ESM). Moreover, we also investigated the individual N-doping patterns, namely pyridinic, pyrrolic, and graphitic N atoms independently incorporated into the carbon lattice, as shown in Fig. S11 (in the ESM). As expected, when nitrogen atoms were independently incorporated onto the carbon surface, the doping facilitated the adsorption of K^+ ions on the carbon surface compared to that of a pure carbon surface. Specifically, pyridinic-N with lone pair electrons and pyrrolic-N with unpaired electrons could bond with K^+ directly with the highest adsorption energies,

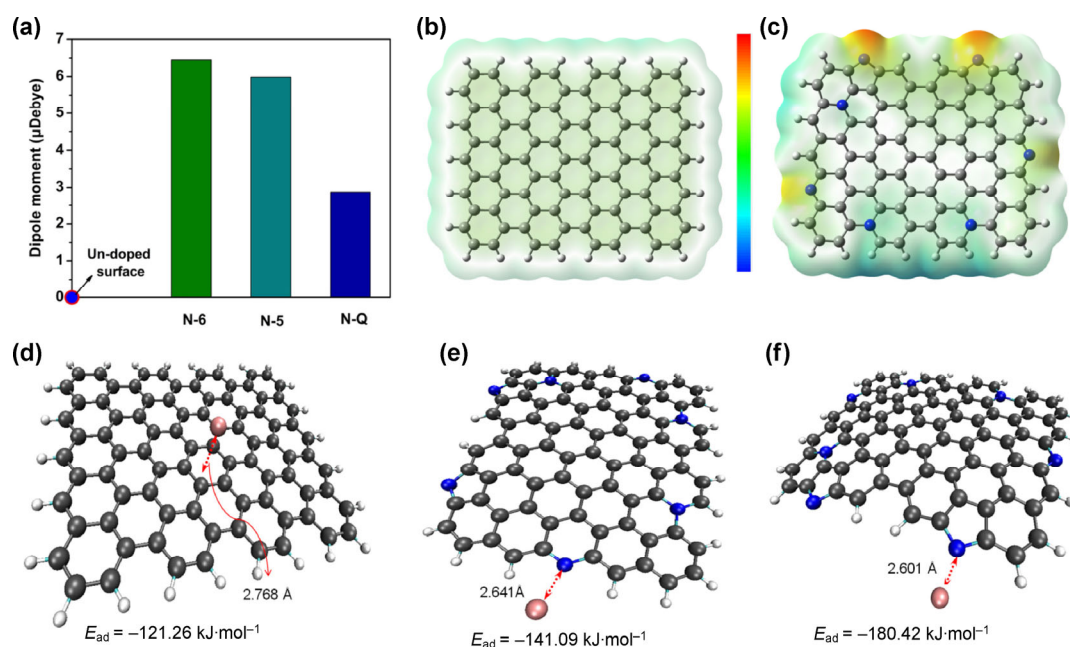


Figure 5 (a) The dipole moments of the non-doped surface and the three surfaces respectively doped with a pyridinic (N-6), pyrrolic (N-5), and graphitic (N-Q) nitrogen atoms. Compared with the non-doped surface without polarization (0.0 Debye), the doped surfaces have dipole moments of 6.45, 5.99, and 2.87 μDebye , respectively, indicating that N-doping effectively induces polarization in the carbon surfaces. For carbon surfaces with multiple doping structures and sites, the effect of polarization effect was visualized by plotting the electrostatic potential onto the surface. (b) and (c) The electrostatic potential of the non-doped carbon surface and the simulated NRC surface, respectively. The electrostatic potential of the non-doped surface is uniform, whereas that of the doped surface contains negative regions, particularly, near areas of concentrated dopants, illustrating the effect of doping.

which potentially provides more sites for ions adsorption and so improves the specific capacitance of EDLC electrodes.

To further reveal the ion-surface interaction mechanism, we calculated the difference electron densities of the pure carbon surface and simulated NRC surface, as illustrated in Fig. 6. These calculations clearly show the interactions between the K^+ ion and the π -electron system [39]. Compared with the adsorption configuration of pure carbon surface, for the simulated NRC surface, we found that the electron density of the pyridinic and the pyrrolic sites decrease while that of the K^+ ion increases, clearly suggesting preferable charge transfer from the doping sites to the K^+ ion. In summary, N doping enhances the transport of electrolyte ions by improving the surface wettability and promoting the ion adsorption onto the doped carbon surface. A combination of these effects increases the effective surface area for charge storage and allows fast transport kinetics for ions and electrons, resulting in the aforementioned improved EDLC performance of the NRC sample.

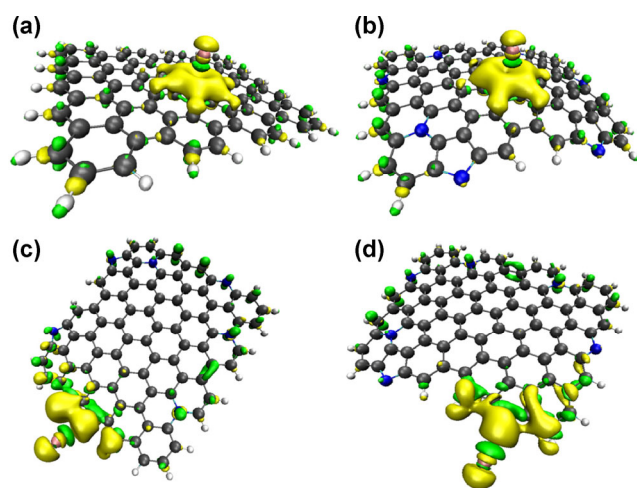


Figure 6 Side views of electron difference density isosurface maps for K^+ adsorption on the non-doped surface and simulated NRC. (a) K^+ ion adsorption above the centroid of the π -system of the undoped surface. (b) K^+ ion adsorption above the centroid of the π -system of the simulated NRC. (c) K^+ ion adsorption on a pyridinic-N site of simulated NRC. (d) K^+ ion adsorption on a pyrrolic N site of simulated NRC. The yellow and green isosurfaces ($\Delta\rho$ of +0.001 to -0.001 a.u., respectively) represent the region in which electron density increases and decreases after K^+ ion adsorption, respectively. $\Delta\rho$ is the differences of electron density before and after adsorption.

3.4 Construction of supercapacitors in aqueous and organic electrolytes

Overall, N-doping increases the polarization of the carbon surface, electrolyte/electrode interactions, and electrolyte wettability, leading to enhanced capacitance, reduced series resistance, and improved rate performance. In addition, N-doping can reduce the Fermi level of the carbon lattice and improve the electronic conductivity [16, 18, 24, 34], factors beneficial in improving the rate performance. To confirm these improvements, we constructed symmetric cells using the NRC particles in 6 M KOH electrolytes. Figures S12(a) and S12(b) in the ESM show the CV curves and galvanostatic charge-discharge curves of the supercapacitor, which shows typical EDLC behavior. As shown in the Ragone plot (Fig. S12(c) in the ESM), the device has an energy density of 9.97 and 4.88 Wh·kg⁻¹ at power densities of 0.125 and 26.23 kW·kg⁻¹, respectively. Such high energy and power densities are among the highest level of currently reported carbon-based symmetric supercapacitors using aqueous electrolytes [5, 18, 40, 41].

To increase the energy density, we also constructed symmetric devices using commercial organic electrolytes (1 M TEABF₄-PC). Figures 7(a) and 7(b) show the CV curves and galvanostatic charge-discharge curves of the electrodes at different current densities. Furthermore, the electrodes could be stably operated at 3.0 V without obvious polarization and IR drop. The specific capacitances calculated from the discharge curves are 162 and 103 F·g⁻¹ at current densities 0.5 and 10 A·g⁻¹, respectively. Figure 7(c) shows the Ragone plot of the NRC capacitor, as well as its comparison with the representative porous carbon-based supercapacitors in a similar electrolyte (TEABF₄-acetonitrile/TEABF₄-PC) [41–44]. The NRC-based supercapacitor can deliver an energy density up to 50.6 Wh·kg⁻¹ at a power density of 0.4 kW·kg⁻¹. Moreover, even at a high power density of 15.6 kW·kg⁻¹, the energy density remains high, at 19.5 Wh·kg⁻¹, demonstrating the excellent rate capability of the cells. Figure 7(d) further shows the cycling stability of the NRC supercapacitor, which maintains ~93% of the initial capacitance after 10,000 cycles at a current density of 2 A·g⁻¹, indicating a long cycling life. Finally, a comparison of our NRC-based

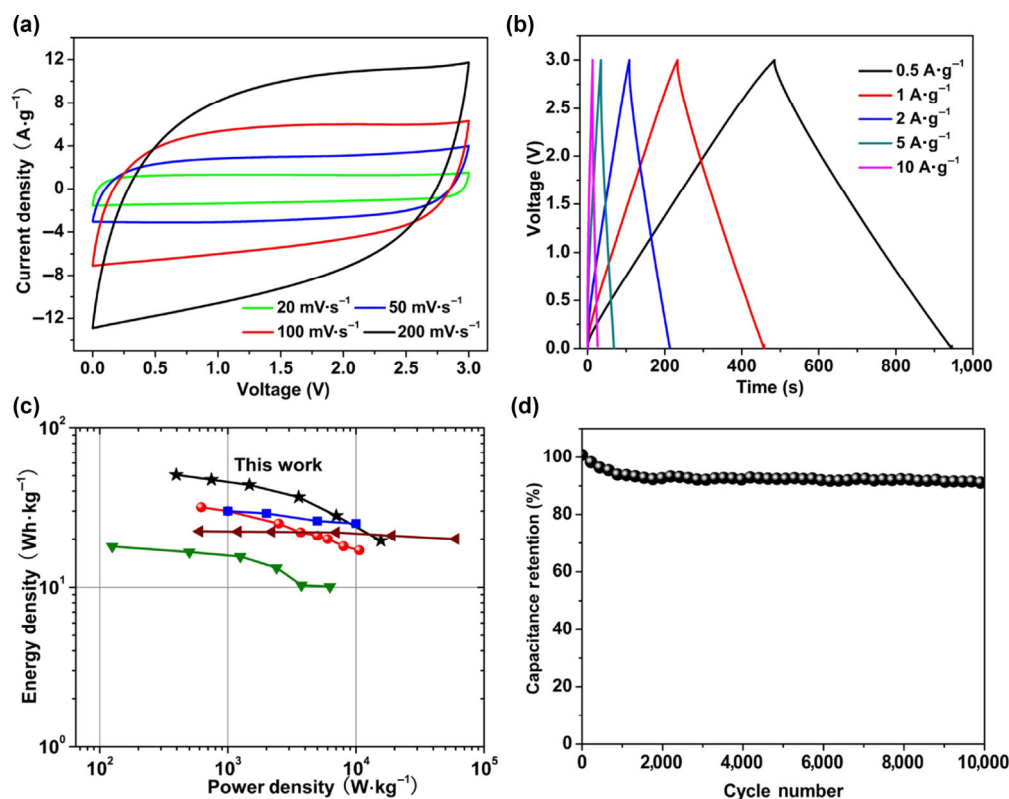


Figure 7 Electrochemical performance of the NRC symmetric supercapacitor in 1 M TEABF₄-PC electrolyte. (a) and (b) CV curves and galvanostatic charge–discharge curves at different scan rates and an operation voltage of 3 V. (c) Ragone plots of the NRC-based capacitor and a comparison with representative symmetric carbon capacitors. (★) The NRC-based capacitor, (●) P-doped hierarchically porous carbon in 1 M TEABF₄-AN electrolyte (Ref. [41]), (■) N-doped graphene oxide in 1M TEABF₄-PC electrolyte (Ref. [44]), (◄) sandwich-type microporous carbon nanosheets in 1 M TEABF₄-AN electrolyte (Ref. [42]), (▼) hierarchical N- and O-enriched porous carbon in 1 M TEABF₄-PC electrolyte (Ref. [43]). (d) Cycling stability of the NRC capacitor operated at current density 2 A·g^{−1} for 10,000 cycles.

supercapacitor with literature reports utilizing the same electrolytes is compiled in Table S3 (in the ESM).

4 Conclusions

In summary, we have used a continuous aerosol-spraying method to synthesize NRC particles with uniform pores, large surface areas, and a high degree of N-doping. Doping with nitrogen enhances the electrolyte/electrode interactions and electrolyte wettability, leading to improved capacitance and rate performance. The combination of a large degree of N-doping, favorable pore structure, and high surface area endows the supercapacitors with high energy and power densities, as well as long cycling lifetimes. This work provides a fundamental basis to understand the enhanced capacitive performance of carbon

materials on heteroatom doping; it also provides a scalable route enabling large-scale production of NRC particles for high-performance supercapacitors.

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Electronic Supplementary Material: Supplementary material (XRD, N₂ adsorption, TGA and supercapacitive characterization results of prepared samples, N-doping carbon surface model and DFT calculation results, and comparison table with reported carbon nanomaterials for symmetric supercapacitors in literature) is available in the online version of this article at <http://dx.doi.org/10.1007/s12274-016-1199-2>.

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